

Business Requirements & Problem Statement Document

Project Title: RideOps — Driver Retention & Operations Intelligence

Target Platform: Pathao Rides (ConsumerTech / Operations)

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Date: September 20, 2026

Document Version: 1.0

1. Executive Summary

In ride-hailing platforms, platform health and ride fulfillment directly depend on maintaining a stable, engaged, and reliable driver supply network. Driver attrition (churn) directly impacts operational capacity, increases driver acquisition costs, and leads to lower ride completion rates and customer dissatisfaction.

This project establishes an **Operations & Business Intelligence framework** designed to identify operational drivers of churn, detect early indicators of driver engagement decay, evaluate driver cohort performance over time, and empower the Operations Team with actionable, data-backed retention strategies.

2. Business Problem & Context

2.1 Problem Description

Ride-hailing platforms experience significant driver turnover. While traditional analysis focuses on post-hoc churn prediction (identifying drivers after they have already left), operational teams require **early behavioral decay signals** to intervene before a driver permanently offboards.

Key operational challenges faced by the Rides Operations Team:

1. **Lack of Early Decay Detection:** Inability to spot subtle drops in driver activity, monthly income, or performance ratings prior to driver departure.

2. **Unclear Cohort Trajectories:** Limited visibility into whether newly onboarded driver cohorts retain better or worse than historical cohorts over 3, 6, and 12-month horizons.
3. **High-Value Driver Attrition:** Inability to isolate and prioritize intervention for high-performing, high-business-value drivers who show signs of disengagement.
4. **Geographic and Segment Variance:** Unidentified variations in churn rates across cities, education levels, and joining designations.

2.2 Strategic & Operational Objectives

The goal of this BI initiative is to transform raw monthly operational driver logs into a structured intelligence layer that answers critical business questions:

- **Operations Management:** Which driver health segments require immediate intervention, and what are the primary operational drivers of disengagement?
- **Driver Acquisition Team:** How do initial onboarding cohorts perform over 1, 3, 6, and 12 months? Are onboarding quality and initial tier assignments improving retention?
- **Driver Operations Team:** Which specific drivers are currently flagged as "At Risk" based on month-over-month performance deterioration?

3. Key Stakeholders & Requirements Matrix

Stakeholder Role	Primary Need / Question	Key Metrics / Artifacts Required
Operations Manager	Macro view of fleet stability, active driver headcount, and regional churn disparities.	Overall Retention Rate, Active Driver Count, Churn by City/Grade, Executive Dashboard.

Driver Operations Team	Micro view of individual driver health to target daily operational interventions.	Driver Health Score Index, Monthly Performance Trend (Income/Rating), Retention Action Center.
Driver Acquisition Team	Evaluation of onboarding cohort quality and long-term driver lifecycle survival.	Cohort Retention Heatmap (1, 3, 6, 12 months), Retention Curves by Joining Designation.
BI & Analytics Lead	Standardized SQL data architecture, reliable ETL pipeline, validated data accuracy, and modular dashboards.	PostgreSQL Schema (Star Schema), Data Quality Audits, SQL Analytical Views, BRD & Documentation.

4. Key Performance Indicators (KPIs) & Logic Framework

4.1 Core Operational Metrics

- **Active Drivers (N_{active}):** Total unique drivers reporting active status/income in a given reporting month.
- **Churned Drivers (N_{churn}):** Drivers with a populated LastWorkingDate within or preceding the reporting period.
- **Monthly Driver Attrition Rate (%):**

$$\text{Attrition Rate}_t = \left(\frac{\text{Drivers Leaving in Month } t}{\text{Total Active Drivers at Start of Month } t} \right) \times 100$$

4.2 Longitudinal & Behavioral Metrics

- **Driver Tenure (Months):** Months elapsed between Dateofjoining and Reporting Date (or LastWorkingDate).
- **Income Delta (ΔIncome):** Percentage change in monthly income compared to the driver's rolling historical baseline:

$$\Delta\text{Income}_t = \left(\frac{\text{Income}_t - \text{Income}_{t-1}}{\text{Income}_{t-1}} \right) \times 100$$

- **Business Value Trend (ΔBV):** Percentage change in monthly revenue generated for the platform.
- **Rating Shift (ΔRating):** Difference between current quarterly rating and prior period rating.

4.3 Composite Index: Driver Health Score

To move from reactive reporting to proactive operational intervention, a rule-based **Driver Health Score (0–100)** is defined:

$$\text{Health Score} = 0.30 \cdot S_{\text{Activity}} + 0.25 \cdot S_{\text{Income}} + 0.20 \cdot S_{\text{BV}} + 0.15 \cdot S_{\text{Rating}} + 0.10 \cdot S_{\text{Tenure}}$$

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Score Bands & Operational Categorization:

- **80 – 100 (Healthy):** Consistent or rising business performance; stable engagement.
- **60 – 79 (Stable):** Normal variance in earnings/activity; low short-term risk.
- **40 – 59 (Watch List):** Minor performance drop ($\geq 15\%$ drop in income or rating decay); candidate for check-in.

- **o – 39 (At Risk):** Severe decay across multiple metrics ($\geq 30\%$ drop in income/business value); requires high-priority retention action.

5. System & Analytics Architecture

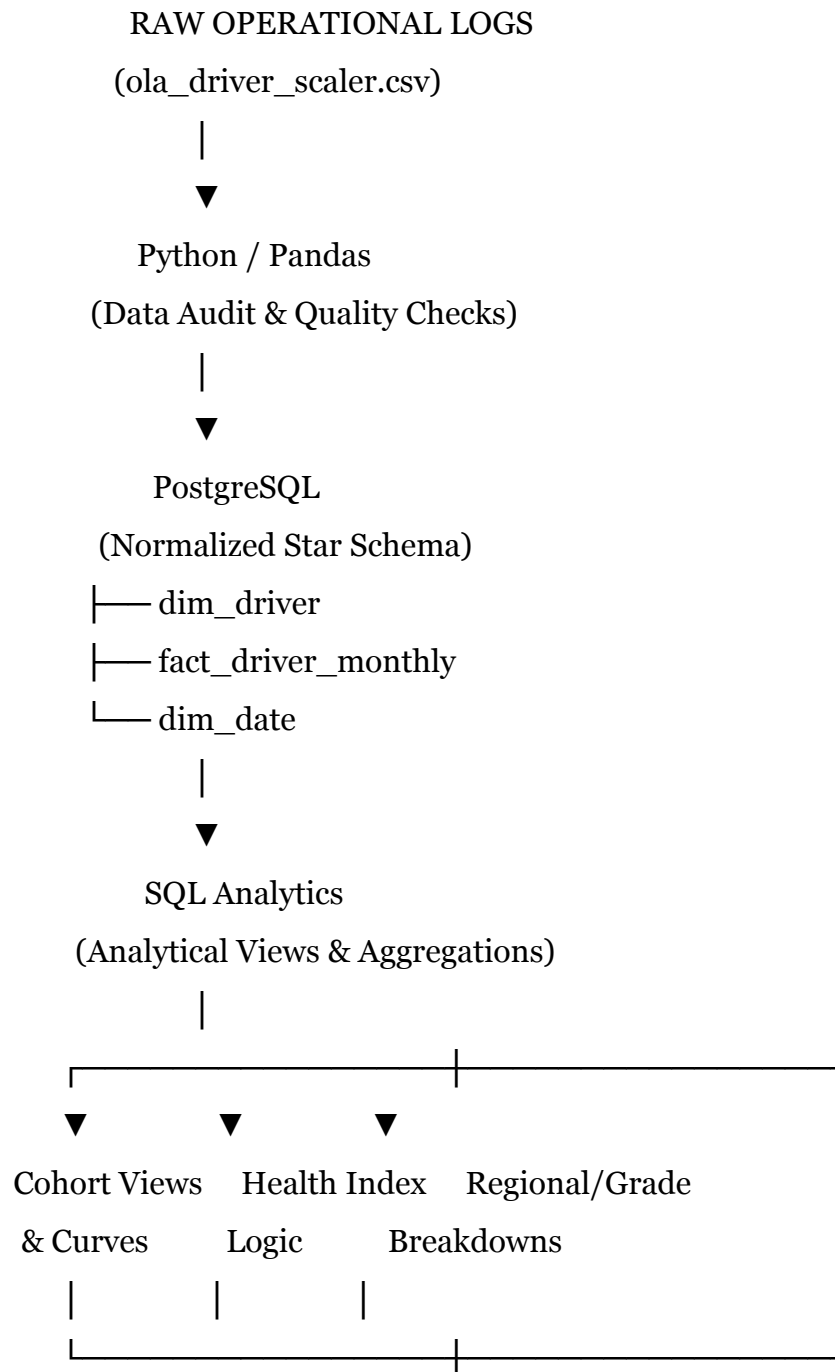




Tableau Public

(Operational Dashboards)

- |— 1. Executive Overview
- |— 2. Driver Retention & Cohorts
- |— 3. Driver Health Deep-Dive
- └— 4. Retention Action Center

6. Scope & Deliverables

Included in Scope

1. **Data Normalization & Modeling:** Designing a PostgreSQL Star Schema (dim_driver, fact_driver_monthly, dim_date).
2. **SQL Analytics Engine:** Production-grade SQL scripts for data quality, cohort retention matrix generation, health score calculation, and executive reporting views.
3. **Statistical Validation:** Python hypothesis testing evaluating whether significant drops in driver income directly correlate with churn.
4. **Tableau Reporting Suite:** Four distinct interactive dashboards catering to Executive, Operational, and Acquisition stakeholders.
5. **Operational Action Center:** Filterable interface identifying high-value drivers showing active performance decay.

7. Business Constraints & Assumptions

1. **Dataset Nature:** The dataset contains monthly snapshots from 2019–2020 covering 2,381 unique drivers and 19,104 monthly observations.
2. **Driver Exit Definition:** Null values in LastWorkingDate represent active drivers; non-null dates represent confirmed driver offboarding.

3. **Data Anonymization:** Data values represent proxy operational logs used to simulate ride-hailing supply dynamics for Pathao Rides.